

Single-Score Safety Radius for the Damped Double Integrator

Detailed Lyapunov and conformal derivation

Scope: One Model, One Calibration Score

- Plant: planar damped double integrator, exactly as used by the repository.
- Reference gap: RRT expert (x_d, u_d) versus LLM prediction $(\hat{x}_d, \hat{u}_d)$.
- Controller: LQR feedback around the LLM reference.
- Certificate: one conformal score for the **combined disturbance entering the error dynamics**.

Target statement

Construct $\rho(t)$ such that, for a fresh exchangeable test environment,

$$\mathbb{P}[\|x(t) - x_d(t)\|_2 \leq \rho(t), \forall t \in [0, T]] \geq 1 - \delta.$$

No nonlinear quadrotor model and no separate q_u, q_x union bound are used.

Damped Double-Integrator Dynamics

State and design-level input:

$$x = [p_x \quad p_y \quad v_x \quad v_y]^T \in \mathbb{R}^4, \quad u = [u_x \quad u_y]^T \in \mathbb{R}^2.$$

With damping $d = 1.1$,

$$\dot{p} = v, \quad \dot{v} = -d v + d u, \quad \dot{x} = Ax + Bu,$$

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1.1 & 0 \\ 0 & 0 & 0 & -1.1 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1.1 & 0 \\ 0 & 1.1 \end{bmatrix}.$$

The model assumes the low-level flight stack realizes the commanded planar input u .

Expert, LLM Reference, and Implementable Controller

Expert and predicted reference dynamics:

$$\dot{x}_d = Ax_d + Bu_d, \quad \dot{\hat{x}}_d = A\hat{x}_d + B\hat{u}_d.$$

Define true tracking error and prediction gaps:

$$e := x - x_d, \quad q_{quadd_x} := \hat{x}_d - x_d, \quad q_{quadd_u} := \hat{u}_d - u_d.$$

The implementable controller tracks the available LLM reference:

$$\hat{e} := x - \hat{x}_d = e - d_x, \quad u = -K\hat{e} + \hat{u}_d = -K(e - d_x) + \hat{u}_d.$$

Important

x_d and u_d are required during calibration, but they are not required online.

Error Dynamics: Combine the Two Gaps First

Starting with $e = x - x_d$,

$$\begin{aligned}\dot{e} &= Ax + Bu - (Ax_d + Bu_d) \\ &= Ae + B[-K(e - d_x) + \hat{u}_d - u_d] \\ &= (A - BK)e + Bd_u + BKd_x.\end{aligned}$$

Define

$$A_c := A - BK, \quad \boxed{w(t) := Bd_u(t) + BKd_x(t)}.$$

Then the entire uncertain error system is

$$\boxed{\dot{e} = A_c e + w.}$$

The closed loop sees only w ; it does not see separate worst-case values of d_u and d_x .

Lyapunov Metric and Certified Decay Rate

Choose $P = P^\top \succ 0$ and

$$V(e) := e^\top P e.$$

The required certificate condition is

$$A_c^\top P + P A_c \preceq -2\alpha_P P, \quad \alpha_P > 0.$$

For fixed P, K , the largest valid rate is

$$\alpha_P = \frac{1}{2} \lambda_{\min} \left(P^{-1/2} \left[-A_c^\top P - P A_c \right] P^{-1/2} \right).$$

Do not substitute the spectral abscissa automatically

– $\max \operatorname{Re} \lambda(A_c)$ describes modal decay, but it need not satisfy the matrix inequality in the chosen P -metric.

Differentiate the Lyapunov Function

Using $\dot{e} = A_c e + w$,

$$\begin{aligned}\dot{V} &= \dot{e}^\top P e + e^\top P \dot{e} \\ &= e^\top (A_c^\top P + P A_c) e + 2e^\top P w \\ &\leq -2\alpha_P V + 2e^\top P w.\end{aligned}$$

Factor $P = P^{1/2} P^{1/2}$ and apply Cauchy–Schwarz:

$$\begin{aligned}e^\top P w &= (P^{1/2} e)^\top (P^{1/2} w) \\ &\leq \|P^{1/2} e\|_2 \|P^{1/2} w\|_2 \\ &= \sqrt{V} \sqrt{w^\top P w}.\end{aligned}$$

Therefore

$$\dot{V} \leq -2\alpha_P V + 2\sqrt{V} \sqrt{w^\top P w}.$$

Scalar Comparison System

Let

$$W := \sqrt{V} = \sqrt{e^\top P e}.$$

For $W > 0$, $\dot{W} = \dot{V}/(2W)$, hence

$$\dot{W} \leq -\alpha_P W + \sqrt{w^\top P w}.$$

If a trajectory satisfies

$$\sup_{t \in [0, T]} \sqrt{w(t)^\top P w(t)} \leq q_w,$$

then

$$\dot{W} \leq -\alpha_P W + q_w.$$

Solving this scalar differential inequality gives

$$W(t) \leq W(0)e^{-\alpha_P t} + \frac{q_w}{\alpha_P}(1 - e^{-\alpha_P t}).$$

One Conformal Nonconformity Score

For calibration environment i , compute the full trajectories and form

$$w_i(t) = B(\hat{u}_{d,i}(t) - u_{d,i}(t)) + BK(\hat{x}_{d,i}(t) - x_{d,i}(t)).$$

Use one scalar score:

$$S_i := \max_{t \in [0, T_i]} \sqrt{w_i(t)^\top P w_i(t)}.$$

Given N exchangeable calibration scores, set

$$k = \lceil (1 - \delta)(N + 1) \rceil, \quad q_w = S_{(\min\{k, N\})},$$

where $S_{(j)}$ is the j th order statistic.

Then

$$\mathbb{P}(S_{\text{test}} \leq q_w) \geq 1 - \delta.$$

Because S_i already maximizes over time, this is a simultaneous trajectory-level event.

Convert the Metric Bound to a State-Norm Tube

Eigenvalue bounds give

$$\sqrt{\lambda_{\min}(P)} \|e\|_2 \leq \sqrt{e^\top P e} = W \leq \sqrt{\lambda_{\max}(P)} \|e\|_2.$$

Thus

$$\rho(t) = \sqrt{\frac{\lambda_{\max}(P)}{\lambda_{\min}(P)}} \|e(0)\|_2 e^{-\alpha_P t} + \frac{q_w}{\alpha_P \sqrt{\lambda_{\min}(P)}} (1 - e^{-\alpha_P t}).$$

With probability at least $1 - \delta$,

$$\|x(t) - x_d(t)\|_2 \leq \rho(t), \quad \forall t \in [0, T].$$

As $t \rightarrow \infty$,

$$\rho_\infty = \frac{q_w}{\alpha_P \sqrt{\lambda_{\min}(P)}}.$$

Numerical CARE Solution Used in the Repository

The current LQR weights are

$$Q = \text{diag}(10, 10, 1, 1), \quad R = I_2.$$

The CARE solution is

$$P = \begin{bmatrix} 8.8032 & 0 & 2.8748 & 0 \\ 0 & 8.8032 & 0 & 2.8748 \\ 2.8748 & 0 & 1.6216 & 0 \\ 0 & 2.8748 & 0 & 1.6216 \end{bmatrix},$$

$$K = \begin{bmatrix} 3.1623 & 0 & 1.7838 & 0 \\ 0 & 3.1623 & 0 & 1.7838 \end{bmatrix}.$$

$$\lambda_{\min}(P) = 0.612622, \quad \lambda_{\max}(P) = 9.812202,$$

$$\sqrt{\kappa(P)} = 4.002092.$$

Certified Rate Versus Eigenvalue Rate

Closed-loop eigenvalues:

$$\lambda(A - BK) = -1.531095 \pm 1.065013j \quad (\text{each repeated twice}).$$

The code currently reports the spectral value

$$\alpha_{\text{eig}} = 1.531095.$$

But the generalized-eigenvalue calculation required by the P -metric gives

$$\alpha_P = 1.096230.$$

Indeed,

$$A_c^\top P + PA_c + 2\alpha_{\text{eig}}P \not\leq 0,$$

whereas

$$A_c^\top P + PA_c + 2\alpha_P P \preceq 0.$$

The radius calculation must use α_P unless a different valid transient bound is proved.

2,000-Sample Single-Score Calibration

Calibration source:

- $N = 2000$ actual refined and verified RRT/LLM prediction pairs;
- shared-clock minimum-control QP trajectories;
- score $S_i = \max_t \sqrt{w_i^\top P w_i}$;
- desired trajectory-level coverage $1 - \delta = 0.90$.

Finite-sample rank:

$$k = \lceil 0.9(2000 + 1) \rceil = \lceil 1800.9 \rceil = 1801.$$

The empirical conformal quantile is

$$q_w = S_{(1801)} = 1.641463.$$

Only one failure budget $\delta = 0.10$ is used; no $\delta_u + \delta_x$ split is required.

Numerical Radius

Insert

$$\begin{aligned}q_w &= 1.641463, & \alpha_P &= 1.096230, \\ \lambda_{\min}(P) &= 0.612622, & \sqrt{\kappa(P)} &= 4.002092.\end{aligned}$$

Steady-state term:

$$\begin{aligned}\rho_\infty &= \frac{q_w}{\alpha_P \sqrt{\lambda_{\min}(P)}} \\ &= \frac{1.641463}{1.096230 \sqrt{0.612622}} \\ &= \boxed{1.913079}.\end{aligned}$$

Complete time-dependent tube:

$$\begin{aligned}\rho(t) &= 4.002092 \|e(0)\|_2 e^{-1.096230t} \\ &\quad + 1.913079(1 - e^{-1.096230t}).\end{aligned}$$

If $e(0) = 0$, the tube grows monotonically from zero toward 1.913079.

Why One Score Is Tighter

The previous two-score construction first applies

$$\|Bd_u + BKd_x\| \leq \|B\|\|d_u\| + \|BK\|\|d_x\|,$$

then combines two separate conformal events with a union bound.

The one-score construction calibrates

$$\sqrt{(Bd_u + BKd_x)^\top P(Bd_u + BKd_x)}$$

directly. It retains:

- the directions of d_u and d_x ;
- their correlation within each trajectory;
- cancellation or reinforcement through B and BK ;
- the exact certificate metric P .

The smaller radius comes from removing norm and union-bound conservatism, not from weakening 90% coverage.

What Is Fixed and What Must Be Recalibrated

Fixed for one certificate

- dynamics matrices A, B ;
- controller gain K ;
- metric P ;
- certified rate α_P ;
- score definition and coverage $1 - \delta$.

Requires recalibration

- changing K or LQR Q, R ;
- changing P or state normalization;
- changing trajectory generation;
- changing clock alignment;
- changing the operating distribution.

Reason

The score contains both P and K through $\sqrt{w^\top P w}$ and $w = Bd_u + BKd_x$.

Interpretation and Assumptions

- The conformal guarantee requires calibration and test environments to be exchangeable.
- The linear model and idealized low-level realization must remain valid over the certified operation.
- The state norm mixes position and velocity components; 1.913079 is a full-state Euclidean bound, not automatically a pure geometric distance.
- Since $\|p - p_d\|_2 \leq \|x - x_d\|_2$, it is numerically conservative for position. A unit-consistent position certificate should use explicit state normalization or output projection.
- This bounds RRT-LLM discrepancy entering the error dynamics; it does not certify obstacle clearance by itself.

Final 90% single-score certificate

$$\mathbb{P}[\|x(t) - x_d(t)\|_2 \leq 4.002092\|e(0)\|e^{-1.096230t} + 1.913079(1 - e^{-1.096230t}), \forall t] \geq 0.90.$$